

Section 1: System Inventory

- Complete table with your actual system specifications:

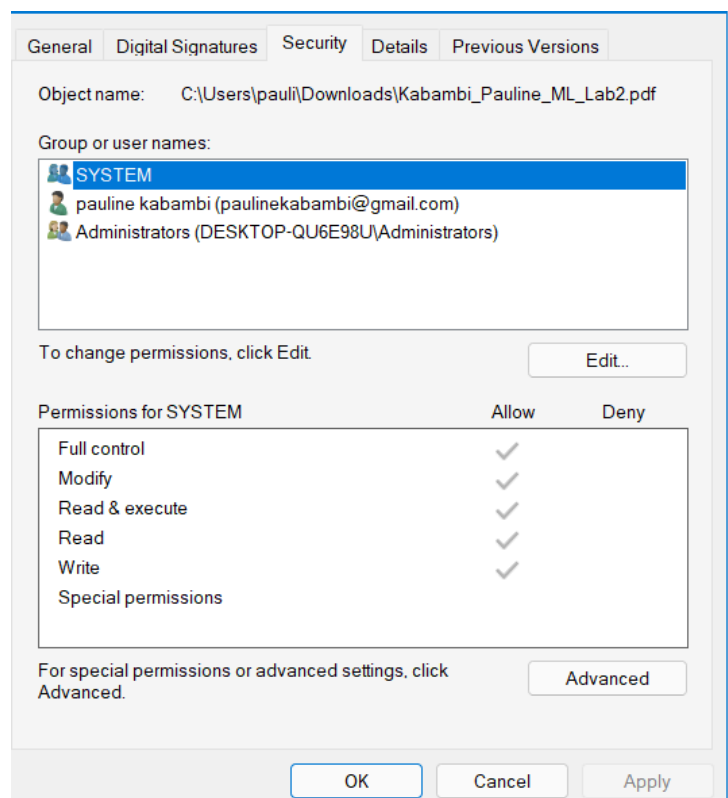
OS name, edition, and full version number	• Windows 11 • 11th Generation • 10.0.26200 Build 26200
Processor details	11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz (2.42 GHz)
RAM Amount (in GB)	8 GB
Storage capacity and type	238 GB / SSD (RAID)

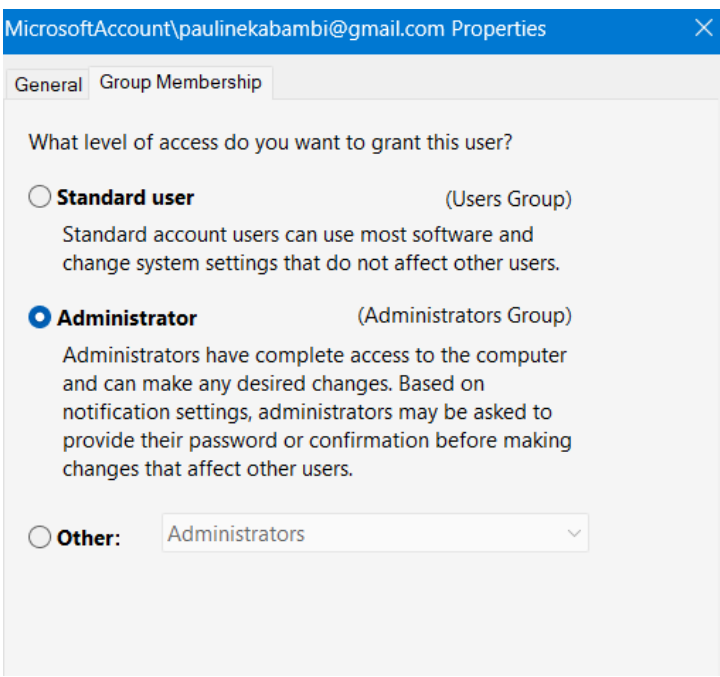
Section 2: Access Control Model

Identification of your OS's access control model (DAC/MAC/RBAC)

My system used the DAC (Discretionary Access Control), as shown in the next box. It is a DAC system because. I can perform the following actions:

- can modify permissions.
- Users with permission can grant access to others.
- Control is discretionary (left to the owner).



Definition cited from official sources (NIST, Microsoft, Red Hat - NOT Wikipedia)	DAC (or Discretionary Access Control) is a type of access control. According to microsoft.com, it is any object that a system (or computer) owner protects and can grant access to users at their discretion. DAC offers case-by-case control over resources.
Real example from YOUR system demonstrating the Principle of Least Privilege My computer's default setting is shown in the next box.	 The screenshot shows a Windows User Account Control dialog box titled "MicrosoftAccount\paulinekabambi@gmail.com Properties". It has two tabs: "General" and "Group Membership". The "Group Membership" tab is active. It asks "What level of access do you want to grant this user?". There are three radio button options: "Standard user" (Users Group), "Administrator" (Administrators Group), and "Other:". The "Administrator" option is selected. Below "Other:" is a dropdown menu showing "Administrators".

Explanation of how your example shows PoLP

As the screenshots above show, on my Windows system, my account (**MicrosoftAccount\paulinekabambi@gmail.com**) is a member of the Administrators group by default. In this specific case, Windows enforces the privacy policy via User Account Control (UAC). Even as an administrator, I don't automatically have elevated privileges. For actions such as installing software or changing system settings, the system always requires explicit permission for privilege escalation. This demonstrates that administrator privileges are restricted until explicit authorization is granted, thus reducing the risk of unauthorized system-wide changes.

Section 3: Process Analysis

For **three different running processes** from your system:

- Process name:
- Process ID (PID)

- Description of what it does
- Security risk hypothesis using attacker perspective (specific risks like privilege escalation, data leakage, lateral movement, DoS)

1) Process 1: Spotify

Spotify.exe	4548	Running	pauli	00	26,732 K	x64	Spotify
Spotify.exe	7300	Running	pauli	00	84 K	x64	Spotify
Spotify.exe	12488	Running	pauli	00	3,236 K	x64	Spotify
Spotify.exe	21484	Running	pauli	00	6,504 K	x64	Spotify
Spotify.exe	11616	Running	pauli	00	1,176 K	x64	Spotify
Spotify.exe	20500	Running	pauli	01	134,768 K	x64	Spotify
Spotify.exe	14700	Running	pauli	00	672 K	x64	Spotify

I noticed that my Spotify has been assigned seven (7) PID numbers. I rarely use Spotify on my computer; however, when I turn on my computer, it is one of the apps that opens automatically. Having the Spotify desktop app on my computer can pose (lower) security risks. An attacker could find a way to log into my account via credential stuffing, potentially steal personal information (email, birthday, payment details), or sell the account on the dark web.

2) Microsoft Edge

msedge.exe	26864	Running	pauli	00	0 K	x64	Microsoft Edge
msedge.exe	36836	Running	pauli	00	4,032 K	x64	Microsoft Edge
msedge.exe	27616	Running	pauli	00	170,864 K	x64	Microsoft Edge
msedge.exe	24952	Running	pauli	00	8,344 K	x64	Microsoft Edge
msedge.exe	27276	Running	pauli	00	540,304 K	x64	Microsoft Edge
msedge.exe	2808	Running	pauli	00	53,360 K	x64	Microsoft Edge
msedge.exe	27304	Running	pauli	00	52,884 K	x64	Microsoft Edge
msedge.exe	27716	Running	pauli	00	31,368 K	x64	Microsoft Edge
msedge.exe	12320	Running	pauli	00	28,128 K	x64	Microsoft Edge
msedge.exe	19304	Running	pauli	00	43,376 K	x64	Microsoft Edge
msedge.exe	28428	Running	pauli	00	31,624 K	x64	Microsoft Edge
msedge.exe	17012	Running	pauli	00	133,212 K	x64	Microsoft Edge
msedge.exe	12508	Running	pauli	00	524 K	x64	Microsoft Edge
msedge.exe	17592	Running	pauli	00	24,004 K	x64	Microsoft Edge
msedge.exe	17600	Running	pauli	00	142,008 K	x64	Microsoft Edge
msedge.exe	17684	Running	pauli	00	4,224 K	x64	Microsoft Edge
msedge.exe	18244	Running	pauli	00	16,048 K	x64	Microsoft Edge
msedge.exe	18352	Running	pauli	00	17,604 K	x64	Microsoft Edge
msedge.exe	13392	Running	pauli	00	1,808 K	x64	Microsoft Edge
msedge.exe	21252	Running	pauli	00	1,120 K	x64	Microsoft Edge
msedge.exe	7380	Running	pauli	00	1,816 K	x64	Microsoft Edge
msedge.exe	21536	Running	pauli	00	243,472 K	x64	Microsoft Edge
msedge.exe	26516	Running	pauli	00	24,328 K	x64	Microsoft Edge

I often used Microsoft to access my Marymount University accounts since Canvas can't be accessed with the Google Chrome browser. I also sometimes used it for personal purposes, but not as much as with Google Chrome. I had to do some research to understand how much vulnerability I am exposed to by using Edge. While Microsoft Edge has built-in security (e.g., SmartScreen) to block known malicious sites. In terms

of denial-of-service, if I access the wrong site, it can cause my Edge to experience severe system slowdowns and consume all of my computer's CPU power. It would make the browser unusable. Additionally, if I have my card information saved in the browser, and a hacker gains access to my personal information, they can steal it, which represents a risk of data leakage.

3. Adobe Acrobat

Name ^	PID	Status	User name	CPU	Memory (ac...	Architec...	Description
 Acrobat.exe	26792	Running	pauli	00	880 K	x64	Adobe Acrobat
 Acrobat.exe	27424	Running	pauli	00	3,180 K	x64	Adobe Acrobat

My Adobe Acrobat has been assigned two (2) PID numbers (pictured below). I usually use Acrobat mainly for reading PDFs and signing documents because I use the free subscription option. The security risks may include data leakage and theft, as well as privilege escalation. A hacker can exploit vulnerabilities in JavaScript parsing within Acrobat to access and export sensitive documents from my computer to an external server. In terms of privilege escalation, an attacker may exploit a flaw while Acrobat is running with limited user rights, triggering a crash that allows them to escalate privileges to the Administrator level, which gives them full control over my laptop.

Section 4: Technical Requirements

- Proper Markdown formatting (headings, tables, lists)
- Professional documentation quality
- Descriptive Git commit message (not "done" or "lab 2")
- Successfully pushed to the GitHub repository

General Digital Signatures Security Details Previous Versions

Object name: C:\Users\pauli\Downloads\Kabambi_Pauline_ML_Lab2.pdf

Group or user names:

- SYSTEM
- pauline kabambi (paulinekabambi@gmail.com)
- Administrators (DESKTOP-QU6E98U\Administrators)

To change permissions, click Edit.

Edit...

Permissions for SYSTEM	Allow	Deny
Full control	✓	
Modify	✓	
Read & execute	✓	
Read	✓	
Write	✓	
Special permissions		

For special permissions or advanced settings, click Advanced.

Advanced

OK Cancel Apply

MicrosoftAccount\paulinekabambi@gmail.com Properties

General Group Membership

What level of access do you want to grant this user?

☐ **Standard user** (Users Group)
Standard account users can use most software and change system settings that do not affect other users.

☒ **Administrator** (Administrators Group)
Administrators have complete access to the computer and can make any desired changes. Based on notification settings, administrators may be asked to provide their password or confirmation before making changes that affect other users.

☐ **Other:** Administrators